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CSA0508 – Database Management System

Software Components of DBMS

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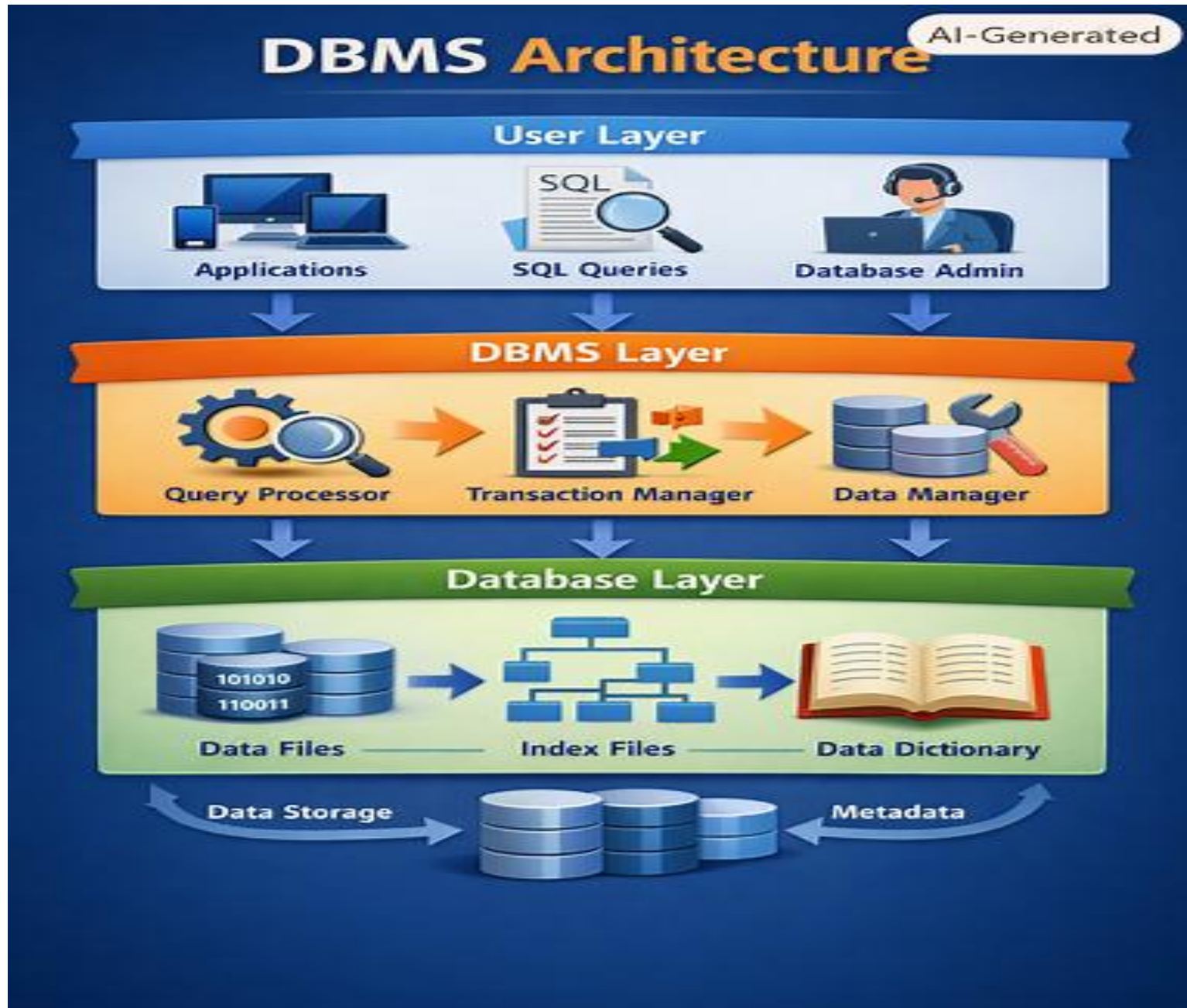
Introduction



- A Database Management System (DBMS) is built from several coordinated software modules.
- The three core internal components are:
 - Query Processor
 - Storage Manager
 - Transaction Manager
- These work together to ensure efficient data retrieval, safe storage, and reliable multi-user access.



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Query Processor



Role

- The query processor interprets user queries (mostly SQL) and converts them into low-level instructions that the DBMS can execute.

Key Functions

- Parsing - Checks SQL syntax and builds a parse tree.
- Query Optimization - Chooses the most efficient way to execute the query using cost-based algorithms.
- Query Execution Engine - Executes the optimized query plan using relational operators (selection, join, projection).



Query Processor



Why It Matters

- Reduces query execution time
- Ensures efficient use of indexes and joins
- Converts high-level SQL into machine-level operations



Query Processor: Parsing & Validation

Key Points:

- SQL query is checked for **syntax errors**
- DBMS builds a **parse tree**
- Validates table names, column names, data types
- Ensures the query follows relational rules

Outcome:

- A structured representation of the query ready for optimization



Query Processor: Query Optimization

Key Points:

- Chooses the **best execution plan**
- Uses **cost-based optimization** (I/O cost, CPU cost, index usage)
- Reorders operations (e.g., push down selections)
- Selects best join method: nested-loop, hash join, sort-merge

Outcome:

- Fastest possible plan for executing the query



Query Processor: Execution Engine



Key Points:

- Uses relational operators:
 - Selection
 - Projection
 - Join
 - Sorting
- Interacts with storage manager to fetch data

Outcome:

- Produces the final result set for the user



Storage Manager



Role:

- The storage manager handles how data is physically stored, organized, and retrieved on disk.

Key Functions:

- **File Manager** Manages files, pages, and blocks on disk.
- **Buffer Manager** Moves data between disk and main memory; reduces disk I/O.
- **Index Manager** Maintains structures like B+ trees and hash indexes for fast access.
- **Record Manager** Handles record formats, layouts, and access methods.



Storage Manager



Why It Matters

- Ensures efficient data storage
- Minimizes disk access time
- Supports indexing and data organization
- Provides data structures for fast retrieval



Storage Manager: File & Page Management



Key Points:

- Organizes data into **files, pages, blocks**
- Handles page allocation, deallocation
- Manages physical placement of data on disk

Outcome:

- Efficient disk usage and structured data storage



Storage Manager: Buffer Manager

Key Points:

- Maintains a **buffer pool** in main memory
- Reduces disk I/O by caching frequently accessed pages
- Uses replacement policies: LRU, MRU, Clock

Outcome:

- Faster query execution due to fewer disk reads



Storage Manager: Index Manager



Key Points:

- Creates and maintains indexes
- Common index types:
 - **B+ Tree**
 - **Hash Index**
- Speeds up search, join, and sorting operations

Outcome:

- Quick access to specific records



Transaction Manager

Role:

- The transaction manager ensures that multiple users can access the database safely without corrupting data.

Key Functions:

- **Concurrency Control** Uses locking, timestamp ordering, or MVCC to prevent conflicts.
- **Recovery Manager** Uses logs and checkpoints to restore the database after crashes.



Transaction Manager



- **Transaction Scheduling** Ensures ACID properties:
- **Atomicity** – all or nothing
- **Consistency** – valid state transitions
- **Isolation** – no interference
- **Durability** – permanent results



Transaction Manager: Concurrency Control

Key Points:

- Prevents conflicts when multiple users access data
- Techniques:
 - **Locking (2PL)**
 - **Timestamp Ordering**
 - **MVCC (Multi-Version Concurrency Control)**

Outcome:

- Ensures isolation and consistency



Transaction Manager: Recovery System

Key Points:

- Maintains **logs** of all operations
- Uses **checkpoints** to reduce recovery time
- After crash:
 - Undo incomplete transactions
 - Redo committed transactions

Outcome:

- Database returns to a consistent state



ACID Properties



- **Atomicity** - All steps of a transaction happen completely or not at all
- **Consistency** - Database moves from one valid state to another
- **Isolation** - Transactions do not interfere with each other
- **Durability** - Committed results survive crashes
- **Outcome** - Reliable and predictable transaction behavior



How These Components Work Together

- The **Query Processor** decides *what* to do.
- The **Storage Manager** decides *where and how* to do it.
- The **Transaction Manager** ensures it is done *safely and reliably*.

Together, they form the backbone of any DBMS.



Conclusion



- A DBMS is not just a data storage tool — it is a complex software system.

The **Query Processor**, **Storage Manager**, and **Transaction Manager** work in coordination to provide:

- Fast query execution
- Efficient data storage
- Reliable multi-user access
- Crash recovery and consistency
- This architecture makes modern databases robust, scalable, and dependable.



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Thank you